

STAT 5200, Notes 4¹

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Readings. Section 4.2.3 in W, which deals with calculation of standard errors of estimates when there is heteroscedasticity. In addition, I'll discuss the hat matrix and leverage, and use of the sandwich package in R. In the next set of notes, I'll discuss Section 12.8 in W, which deals with simulation and resampling. I'll discuss the use of simulation to determine standard errors (when this is possible), and introduce bootstrapping. I'll consider both the parametric bootstrap and the nonparametric bootstrap, and then calculate standard errors in examples via the bootstrap.

Heteroscedasticity. Let's review the discussion in Notes 3 of the asymptotic properties of the OLS estimator $\hat{\beta}$ of β , and let's extend the discussion. We can write

$$\sqrt{N}(\hat{\beta} - \beta) = \left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}'_i \mathbf{x}_i \right)^{-1} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N u_i \mathbf{x}'_i \right). \quad (1)$$

As N tends to infinity, the first factor on the right-hand side converges in probability to $\mathbf{A}^{-1}$, where $\mathbf{A} = E[\mathbf{x}'\mathbf{x}]$, and the second factor converges in distribution, by the central limit theorem to $N(\mathbf{0}, \mathbf{B})$, where ($Var(\cdot)$ denotes covariance matrix)

$$\mathbf{B} = Var(u\mathbf{x}') = E[(u\mathbf{x}' - E[u\mathbf{x}'])(u\mathbf{x} - E[u\mathbf{x}])]. \quad (2)$$

If Assumption 1 in Notes 2 holds, then $E[u\mathbf{x}'] = \mathbf{0}$, and (2) becomes,

$$\mathbf{B} = E[u^2 \mathbf{x}'\mathbf{x}]. \quad (3)$$

Then (1) converges in distribution to

$$N(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}). \quad (4)$$

Thus the asymptotic covariance matrix of the OLS estimator $\hat{\beta}$ is given by an estimate of

$$\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1} / N. \quad (5)$$

A homoscedasticity condition simplifies the form of this asymptotic covariance matrix and results in the *usual standard errors for the OLS estimates*. The condition was stated previously as follows.

¹This is based on Paul Shaman's lecture notes (Notes 4 from STAT 5200) at the University of Pennsylvania. Last revised: Summer, 2025. Last compiled: October 20, 2025. Lecture notes and associated materials are licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0) license.

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Assumption 3. $E[u^2 \mathbf{x}' \mathbf{x}] = \sigma^2 E[\mathbf{x}' \mathbf{x}]$ where $\sigma^2 = E[u^2]$.

If this assumption holds, then $\mathbf{B} = \sigma^2 \mathbf{A}$, and (4) and (5) become

$$N(\mathbf{0}, \sigma^2 \mathbf{A}^{-1}) \quad \text{and} \quad \sigma^2 \mathbf{A}^{-1}/N. \quad (6)$$

Let's consider what is a sufficient condition for Assumption 3. Certainly Assumption 3 is valid if $E[u^2 | \mathbf{x}] = \sigma^2$; this follows immediately by iterated expectations.

Should Heteroscedasticity Be Expected? Suppose the correct model is given by a conditional expectation. That is, $y = E[y | \mathbf{x}] + u^*$. Further, suppose that one uses the linear projection (linear regression) $\mathbf{x}\beta$ to approximate the conditional expectation. Then

$$y - \mathbf{x}\beta = y - E[y | \mathbf{x}] + E[y | \mathbf{x}] - \mathbf{x}\beta = u^* + E[y | \mathbf{x}] - \mathbf{x}\beta,$$

and

$$\begin{aligned} E[(y - \mathbf{x}\beta)^2 | \mathbf{x}] &= \underbrace{\text{Var}(u^* | \mathbf{x})}_{=\text{Var}(y | \mathbf{x})} + 2 \underbrace{E[u^*(E[y | \mathbf{x}] - \mathbf{x}\beta) | \mathbf{x}]}_{=0 \text{ because } E[u^* | \mathbf{x}] = 0} + E[(E[y | \mathbf{x}] - \mathbf{x}\beta)^2 | \mathbf{x}] \\ &= \text{Var}(y | \mathbf{x}) + (E[y | \mathbf{x}] - \mathbf{x}\beta)^2 \end{aligned} \quad (7)$$

This states that there are two potential reasons that heteroscedasticity arises. The first term indicates that the conditional variance of y can differ by x values (e.g., wages (y) are more dispersed for more educated (x) workers). The second term indicates that the residual variance in a linear regression tends to increase as the difference between the true model (the conditional expectation) and the fitted model (the linear projection) increases. That is, homoscedasticity is unlikely – we should expect the presence of heteroscedasticity.

If Assumption 3 (OLS.3) does not hold, we have heteroscedasticity. That is, if $E[u^2 | \mathbf{x}]$ does not equal $E[u^2] = \sigma^2$, $\mathbf{B} = E[u^2 \mathbf{x}' \mathbf{x}]$ does not reduce to $\sigma^2 E[\mathbf{x}' \mathbf{x}]$. The consequence is that we cannot use the standard errors commonly provided by statistical programs. In many situations this is not a problem – one simply uses standard errors obtained from the sandwich expression $\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}$. There are situations, however, where one wants to make an adjustment so that homoscedasticity is induced. For example, in some cases one can transform the response and explanatory variables to achieve this end.

Standard error calculation.

Usual standard errors (assuming homoscedasticity). The usual standard errors employ an estimate of the asymptotic covariance matrix $\sigma^2 \mathbf{A}^{-1}/N$; see (6). This estimate is written as

$$\widehat{\text{Avar}}(\hat{\beta}) = \hat{\sigma}^2 (\mathbf{X}' \mathbf{X})^{-1}, \quad (8)$$

where

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_N \end{pmatrix}.$$

Heteroscedasticity-consistent (robust) standard errors. If there is heteroscedasticity, the usual standard errors do not apply. If a reasonable model for $\text{Var}(y|\mathbf{x})$ can be formulated, one can employ *weighted least squares* for estimation, as discussed in Appendix. To do so, for each i one divides y_i and each element of $\mathbf{x}_i$ (including the constant 1 for the intercept) by an estimate of $[\text{Var}(y_i|\mathbf{x}_i)]^{1/2}$. Then one applies OLS to the weighted data. Of course, this will be effective only if one's model for $\text{Var}(y|\mathbf{x})$ is correct, often questionable. Sometimes repeated observations are available at each $\mathbf{x}$ value and one can estimate each $[\text{Var}(y_i|\mathbf{x})]^{1/2}$.

An alternative to weighted least squares which is generally applicable, and which is now widely employed, is to use an estimate of the asymptotic covariance matrix given in (5). This estimate can be written as

$$\widehat{\text{Avar}}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^N \hat{u}_i^2 \mathbf{x}_i' \mathbf{x}_i \right) (\mathbf{X}'\mathbf{X})^{-1}, \quad (9)$$

where

$$\hat{u}_i = y_i - \mathbf{x}_i' \hat{\beta}$$

is the i th residual. The estimator (9) is termed the *sandwich estimator*. The standard errors of the elements of the estimator $\hat{\beta}$ are the square roots of the diagonal entries of the matrix (9). They are variously referred to as heteroscedasticity-consistent (HC) standard errors, Eicker-White standard errors, Huber-White standard errors, and Eicker-Huber-White standard errors. See Eicker (1967), Huber (1967) and White (1980b) – all in Wooldridge's list of references.

We should note that

$$\frac{1}{N} \sum_{i=1}^N \hat{u}_i^2 \mathbf{x}_i' \mathbf{x}_i \quad (10)$$

is a consistent estimator of $\mathbf{B} = E[u^2 \mathbf{x}' \mathbf{x}]$. The proof is outlined in Problem 4.4 of Wooldridge. Let's examine the proof. First note that the i th residual is

$$\hat{u}_i = y_i - \mathbf{x}_i' \hat{\beta} = \mathbf{x}_i' \beta + u_i - \mathbf{x}_i' \hat{\beta} = u_i - \mathbf{x}_i' (\hat{\beta} - \beta),$$

so that

$$\hat{u}_i^2 = u_i^2 - 2u_i \mathbf{x}_i' (\hat{\beta} - \beta) + [\mathbf{x}_i' (\hat{\beta} - \beta)]^2.$$

Then (10) is

$$\frac{1}{N} \sum_{i=1}^N u_i^2 \mathbf{x}_i' \mathbf{x}_i - \frac{2}{N} \sum_{i=1}^N u_i \mathbf{x}_i' (\hat{\beta} - \beta) \mathbf{x}_i' \mathbf{x}_i + \frac{1}{N} \sum_{i=1}^N \left(\mathbf{x}_i' (\hat{\beta} - \beta) (\hat{\beta} - \beta)' \mathbf{x}_i \right) \mathbf{x}_i' \mathbf{x}_i. \quad (11)$$

Assume that the expectations of all parts of (11) exist and are finite. The expectation of the first sum in (11) is clearly the desired $\mathbf{B} = E[u^2 \mathbf{x}' \mathbf{x}]$. The sample averages given by the other two sums in (11) converge to $\mathbf{0}$ in probability because $\hat{\beta} - \beta = o_p(1)$; that is, $\hat{\beta}$ converges in probability to β , and the sample averages with the remaining product components converge to their expectations, which exist and are finite. Details are in Appendix 2.

Computation and heteroscedasticity options with R. To calculate HC standard errors, use the sandwich package.

```
>install.packages("sandwich")
```

Then make it available with the command

```
>library("sandwich")
```

You should look at the vignette by Achim Zeileis, entitled Econometric Computing with HC and HAC Covariance Matrix Estimators. Go to cran.r-project.org and click on Packages. Then, select sandwich: Robust Covariance Matrix Estimators. The sandwich package is designed for use with both regression and time series applications; we need only the former.

The package supports several options that are all asymptotically valid. These include (9), the suggestion by White (1980b). The other options replace $\hat{u}_i^2$ in (9) by different expressions. Before examining the list of options, let's discuss the hat matrix. For the model we use the notation, as previously,

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_N \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_N \end{pmatrix}$$

so that

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{u}.$$

The projection matrix which produces the vector of predicted values $\hat{\mathbf{y}}$ is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'. \quad (12)$$

The parameter estimator is $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, and the projection of $\mathbf{y}$ is $\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y}$. The matrix $\mathbf{H}$ is called the *hat matrix* (it transforms $\mathbf{y}$ to $\hat{\mathbf{y}}$). The i th diagonal element of $\mathbf{H}$ is h_{ii} , typically written h_i , and these h_i values are called the *hats*. The hats satisfy $0 \leq h_i \leq 1$,

$i = 1, \dots, N$.³ The hat value h_i describes the leverage of the i th data point y_i . It measures in a sense the distance of the i th data point from the remaining $N - 1$ data points in the (K -dimensional) space of the explanatory variables. A data point with a large value of h_i is said to have high leverage. Points which are outliers in the space of the $\mathbf{x}$ observed values have high leverage. These have the potential to greatly influence the determination of estimated slope values in a least squares calculation. I'll discuss some examples of outliers and influential points in class.

The sum of the diagonal elements of $\mathbf{H}$ (the trace of $\mathbf{H}$) is equal to K . Thus the hats add to K . If an observation has high leverage, it essentially has to take leverage away from the other points because of the sum restriction. It is common (but not universal) to view a point as having high leverage if its hat value exceeds twice the average leverage, which is $2K/N$. If h_i is very high, say near 1, then the prediction $\hat{y}_i$ is virtually determined by the value y_i .⁴

Now let's return to discussion of the options in the sandwich package. The package allows six options for the expression in the position occupied by $\hat{u}_i^2$ in formula (9). The options are labeled const, HC0, HC1, HC2, HC3 and HC4 and are as follows:

const	$\hat{\sigma}^2$
HC0	$\hat{u}_i^2$
HC1	$\frac{N}{N-K} \hat{u}_i^2$
HC2	$\frac{\hat{u}_i^2}{1-h_i}$
HC3	$\frac{\hat{u}_i^2}{(1-h_i)^2}$
HC4	$\frac{\hat{u}_i^2}{(1-h_i)^{\delta_i}}$, where $\delta_i = \min(4, h_i/\bar{h})$

For HC4 $\bar{h}$ is the average of the hats, which is K/N . The option const assumes the data are homoscedastic, and HC0 gives standard errors with (9). HC1, HC2 and HC3 were introduced by MacKinnon and White (1985) to improve small sample performance. HC4 was proposed by Cribari-Neto (2004), *Computational Statistics and Data Analysis* **45** 215-233. The spirit of the options HC2,

³The proof uses two results from linear algebra. Let us call them Fact 1 and Fact 2. [Fact 1]: $\mathbf{H}$ is symmetric idempotent, and thus all its eigenvalues are equal to 0 or 1. This is because we can write (Note also that $\mathbf{H}^2 = \mathbf{H}$)

$$\mathbf{A} = \mathbf{U}\mathbf{H}\mathbf{U}' = \mathbf{U}\mathbf{H}^2\mathbf{U}' = \mathbf{U}\mathbf{H}\mathbf{U}'\mathbf{U}\mathbf{H}\mathbf{U}' = \mathbf{A}^2,$$

where $\mathbf{A}$ is the diagonal matrix with eigenvalues of $\mathbf{H}$ and $\mathbf{U}$ is the orthogonal matrix which diagonalizes $\mathbf{H}$. Diagonal elements in $\mathbf{A}$ have to be either 0 or 1. [Fact 2]: For any $N \times 1$ column vector $\mathbf{a}$ normalized so that $\mathbf{a}'\mathbf{a} = 1$, the quadratic form $\mathbf{a}'\mathbf{H}\mathbf{a}$ is bounded between the minimum and maximum eigenvalues of $\mathbf{H}$. Putting Fact 1 and Fact 2 together (letting $\mathbf{a}$ be a vector of zeros except i th element being 1), h_i should be between 0 and 1. In fact, the minimum hat value is $1/N$, which we haven't proved yet.

⁴(Univariate case with the intercept and a single predictor, x_i) If one estimates via $L(y|1, x)$, the hats are given by

$$h_i = \frac{1}{N} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^N (x_j - \bar{x})^2}, \quad i = 1, \dots, N,$$

where $\bar{x}$ is the average of the x_i values.

HC3 and HC4 is to make adjustments for high leverage points in calculating standard errors, rather than exclude the points from the estimation. The adjustments inflate the standard errors, with inflation increasing as leverage increases.

HC3 has a close connection to the jackknife (also, related to the leave-one-out cross-validation) that I will introduce in the next notes. For example, it turns out that the leave-one-out estimator for the mean-squared-error (MSE), $E[u_i^2]$ is

$$\text{leave-one-out cross-validation estimator for MSE} = \frac{1}{N} \sum_{i=1}^N \frac{1}{(1 - h_i)^2} \hat{u}_i^2.$$

which can be also computed by

1. Let observation i^* form the test set, and fit the model using the remaining data. That is, estimate β using observations $i = 1, 2, \dots, i^* - 1, i^* + 1, \dots, N$. Call it $\hat{\beta}_{-i^*}$. Define and compute the prediction error for observation i^* as

$$\hat{u}_{i^*}^* = y_{i^*} - x_{i^*} \hat{\beta}_{-i^*}$$

2. Repeat step 1 for $i^* = 1, 2, \dots, N$.
3. Compute the MSE from $\hat{u}_1^*, \hat{u}_2^*, \dots, \hat{u}_N^*$.

$$\widehat{MSE} = \frac{1}{N} \sum_{i=1}^N (\hat{u}_i^*)^2$$

You can prove that you don't have to do the actual cross-validation, as you can compute it directly from the original residual and hat values:

$$\frac{1}{N} \sum_{i=1}^N (\hat{u}_i^*)^2 = \frac{1}{N} \sum_{i=1}^N \frac{1}{(1 - h_i)^2} \hat{u}_i^2$$

Examples. See [Lab03.pdf](#) for computing HC0, HC1, HC2, HC3, and HC4 using the data.

Appendix 1 (Optional). This appendix⁵ discusses calculation of standard errors in multiple regression. Let's begin by considering simple regression. We assume throughout that there is homoscedasticity.

Let x be a scalar explanatory variable, y a response variable, and assume

$$E[y|x] = \beta_0 + \beta_1 x, \quad y = E[y|x] + u, \quad E[u^2|x] = \sigma^2,$$

with u and x uncorrelated. Assume a sample $(y_i, x_i), i = 1, \dots, N$, is at hand. The standard error of the estimator of the slope attached to x in $L(y/1, x)$ is

$$\frac{\hat{\sigma}}{\sqrt{N-1}sd(x)}, \quad (13)$$

where $\hat{\sigma}^2$ is the usual estimator of σ^2 , given by the sum of squared regression residuals divided by $N-2$, and $sd(x)$ is the standard deviation of the values $x_1, \dots, x_N$. Let's prove (13). The least squares estimator of $\beta = (\beta_0, \beta_1)'$ is $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1)'$. The limiting distribution of $\sqrt{N}(\hat{\beta} - \beta)$ is $N(\mathbf{0}, \sigma^2 \mathbf{A}^{-1})$, where

$$\mathbf{A} = \begin{pmatrix} 1 & E[x] \\ E[x] & E[x^2] \end{pmatrix}$$

Thus, the asymptotic covariance matrix of the estimator $\hat{\beta}$ is $\sigma^2 \mathbf{A}^{-1}/N = \sigma^2 (N\mathbf{A})^{-1}$, and we estimate this by $\hat{\sigma}^2 (N\hat{\mathbf{A}})^{-1}$, where $\hat{\sigma}^2$ is the sum of squared regression residuals divided by $N-2$ and

$$N\hat{\mathbf{A}} = \begin{pmatrix} N & \sum_{i=1}^N x_i \\ \sum_{i=1}^N x_i & \sum_{i=1}^N x_i^2 \end{pmatrix}.$$

The standard error of the slope estimator is the square root of the second diagonal entry of $\hat{\sigma}^2 (N\hat{\mathbf{A}})^{-1}$, which is the square root of

$$\frac{\hat{\sigma}^2}{\sum_{i=1}^N (x_i - \bar{x})^2} = \frac{\hat{\sigma}^2}{(N-1)Var(x)},$$

where $Var(x)$ is the sample variance of the values $x_1, \dots, x_N$.

Now consider multiple regression. Assume

$$E[y|x_1, \dots, x_K] = \beta_0 + \beta_1 x_1 + \dots + \beta_K x_K, \\ y = E[y|x_1, \dots, x_K] + u, \quad E[u^2|x_1, \dots, x_K] = \sigma^2,$$

⁵As one can see, this part of the notes has nothing to do with heteroscedasticity. I will move this part to Notes 3 from the next academic year. The variance inflation factor (VIF) discussed in this section will be discussed in later notes as well.

and that u is uncorrelated with x_k , $k = 1, \dots, K$. We can use Property LP.7 to show that the standard error of the estimator of the partial slope attached to x_k is

$$\frac{\hat{\sigma}\sqrt{VIF(x_k)}}{\sqrt{N-1}sd(x_k)}, \quad (14)$$

where $\hat{\sigma}^2$ is the usual estimator of σ^2 , given by the sum of squared regression residuals from the regression of y on $x_1, \dots, x_K$, divided by $N - K$, $sd(x_k)$ is the standard deviation of the N values of the explanatory variable x_k , and $VIF(x_k)$ is called the variance inflation factor associated with x_k . Let's prove (14) and give an expression for the variance inflation factor.

By Property LP.7 the partial slope attached to x_k in the multiple regression is equal to the slope in the simple regression of v versus r_k , where r_k is the residual from the regression of x_k versus $x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_K$ and v is the residual from the regression of y versus $x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_K$. From the simple regression case above, the standard error of the partial slope estimator is

$$\frac{\hat{\sigma}}{\sqrt{N-1}sd(r_k)}.$$

Rewrite the above as

$$\frac{\hat{\sigma}}{\sqrt{N-1}sd(x_k)} \times \left(\frac{Var(x_k)}{Var(r_k)} \right)^{1/2},$$

where $Var(\cdot)$ denotes a sample variance, and note that

$$\frac{Var(x_k)}{Var(r_k)} = \frac{1}{1 - R_{k \cdot 1, \dots, k-1, k+1, \dots, K}^2},$$

where $R_{k \cdot 1, \dots, k-1, k+1, \dots, K}^2$ is the value of R square from the regression of x_k on $x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_K$. Thus,

$$VIF(x_k) = \frac{1}{1 - R_{k \cdot 1, \dots, k-1, k+1, \dots, K}^2}$$

$VIF(x_k)$ is greater than or equal to 1. It is equal to one if x_k is uncorrelated with all the other x 's, and its value increases as the multiple correlation between x_k and the other x 's increases. The square root of $VIF(x_k)$ specifies the multiplicative inflation attributable to collinearity of the standard error of the partial slope estimate attached to x_k .

Appendix 2. Problem 4.4 in Wooldridge. Let's repeat the problem to be addressed. The expression

$$\frac{1}{N} \sum_{i=1}^N \hat{u}_i^2 \mathbf{x}_i' \mathbf{x}_i \quad (15)$$

is a consistent estimator of $\mathbf{B} = E[u^2 \mathbf{x}' \mathbf{x}]$. Let's examine the proof. First note that the i th residual is

$$\hat{u}_i = y_i - \mathbf{x}_i' \hat{\beta} = \mathbf{x}_i' \beta + u_i - \mathbf{x}_i' \hat{\beta} = u_i - \mathbf{x}_i' (\hat{\beta} - \beta),$$

so that

$$\hat{u}_i^2 = u_i^2 - 2u_i \mathbf{x}_i' (\hat{\beta} - \beta) + [\mathbf{x}_i' (\hat{\beta} - \beta)]^2.$$

Then (15) is

$$\frac{1}{N} \sum_{i=1}^N u_i^2 \mathbf{x}_i' \mathbf{x}_i - \frac{2}{N} \sum_{i=1}^N u_i \mathbf{x}_i' (\hat{\beta} - \beta) \mathbf{x}_i' \mathbf{x}_i + \frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i' (\hat{\beta} - \beta) (\hat{\beta} - \beta)' \mathbf{x}_i) \mathbf{x}_i' \mathbf{x}_i \quad (16)$$

By the Law of Large Numbers, the first term in (16) converges in probability to $\mathbf{B}$ and we need to argue that the other two terms in (16) converge in probability to $\mathbf{0}$.

We first analyze the second term. Let $\mathbf{x}_{il}$ be the l -th element of $\mathbf{x}_i$ and β_l be the l -th element of β .

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^N \underbrace{(u_i \mathbf{x}_i' (\hat{\beta} - \beta))}_{1 \times 1} \underbrace{\mathbf{x}_i' \mathbf{x}_i}_{K \times K} &= \frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i' \mathbf{x}_i) (u_i \mathbf{x}_i' (\hat{\beta} - \beta)) \quad \text{move scalar from left to right} \\ &= \frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i' \mathbf{x}_i) \sum_{l=1}^K u_i \mathbf{x}_{il} (\hat{\beta}_l - \beta_l) \quad \text{from inner product to sum over } l \\ &= \sum_{l=1}^K \left(\underbrace{\left[\frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i' \mathbf{x}_i) u_i \mathbf{x}_{il} \right]}_{\rightarrow_p E[\mathbf{x}_i' \mathbf{x}_i u_i \mathbf{x}_{il}]} \underbrace{(\hat{\beta}_l - \beta_l)}_{\rightarrow_p 0} \right) \quad \text{move "sum over } l \text{" outside} \end{aligned}$$

Each element in $\sum_{l=1}^K (\cdot)$ behaves as $O_p(1) o_p(1)$. As a result, the entire sum behaves as

$$O_p(1) o_p(1) + O_p(1) o_p(1) + \dots + O_p(1) o_p(1) = o_p(1).$$

The similar logic applies to the third term and it behaves as $o_p(1)$ as well. This is because

$$\underbrace{(\mathbf{x}_i' (\hat{\beta} - \beta) (\hat{\beta} - \beta)' \mathbf{x}_i)}_{1 \times 1} = \left(\sum_{l=1}^K \mathbf{x}_{il} (\hat{\beta}_l - \beta_l) \right) \left(\sum_{l'=1}^K \mathbf{x}_{il'} (\hat{\beta}_{l'} - \beta_{l'}) \right) = \sum_{l=1}^K \sum_{l'=1}^K x_{il} x_{il'} (\hat{\beta}_l - \beta_l) (\hat{\beta}_{l'} - \beta_{l'})$$

and therefore

$$\frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i' (\hat{\beta} - \beta) (\hat{\beta} - \beta)' \mathbf{x}_i) \mathbf{x}_i' \mathbf{x}_i = \sum_{l=1}^K \sum_{l'=1}^K \left(\underbrace{\left[\frac{1}{N} \sum_{i=1}^N x_{il} x_{il'} \right]}_{\rightarrow_p E[x_{il} x_{il'}]} \underbrace{(\hat{\beta}_l - \beta_l)}_{\rightarrow_p 0} \underbrace{(\hat{\beta}_{l'} - \beta_{l'})}_{\rightarrow_p 0} \right) = o_p(1).$$

Appendix 3: Weighted Least Squares. Weighted least squares is a methodology that is widely used, often with the aim of inducing homoscedasticity. But there are other purposes for its use. Consider the model $y = \mathbf{x}\beta + u$, and suppose that

$$E[y|\mathbf{x}] = \mathbf{x}\beta \quad \text{and} \quad E[u^2|\mathbf{x}] = \text{Var}(y|\mathbf{x}).$$

Then,

$$y^* = \mathbf{x}^*\beta + u^*,$$

where

$$y^* = y/\sqrt{\text{Var}(y|\mathbf{x})}, \quad \mathbf{x}^* = \mathbf{x}/\sqrt{\text{Var}(y|\mathbf{x})}, \quad u^* = u/\sqrt{\text{Var}(y|\mathbf{x})},$$

is a revision of the model with homoscedasticity incorporated, and the least squares solution for the parameter vector in this formulation is obtained from minimization with respect to β of the expression

$$E[(y^* - \mathbf{x}^*\beta)^2] = E[(y - \mathbf{x}\beta)^2 / \text{Var}(y|\mathbf{x})].$$

The result is

$$\beta_{WLS} = [E[(\mathbf{x}^{*'}\mathbf{x}^*)]]^{-1} E[\mathbf{x}^{*'}y^*],$$

with the subscript attached to emphasize that the weighted least squares solution is usually different from the OLS solution

$$\beta_{OLS} = [E[\mathbf{x}'\mathbf{x}]]^{-1} E[\mathbf{x}'y].$$

With a random sample of data $\{(y_i, \mathbf{x}_i) : i = 1, \dots, N\}$, one performs weighted least squares by minimizing with respect to β the expression

$$\sum_{i=1}^N (y_i - \mathbf{x}_i\beta)^2 w(\mathbf{x}_i),$$

where $w(\mathbf{x}_i) = 1/\text{Var}(y|\mathbf{x}_i)$. Of course, this requires knowledge of the conditional variance of y for each value $\mathbf{x}_i$, and such knowledge is usually not available. If there are multiple values of y for each value of $\mathbf{x}$, one can estimate the values $\text{Var}(y|\mathbf{x})$. However, such estimates may not be very reliable, especially if there are only a few values of y for some values of $\mathbf{x}$. If this approach is reliable, though, one can use the usual standard errors stemming from a homoscedasticity assumption.

Sometimes the values of $\text{Var}(y|\mathbf{x})$ are known up to a scalar constant. For example, suppose that, instead of microdata, you have y averages grouped at the level of the $\mathbf{x}$ values. This typically occurs when the original data violate confidentiality of subjects. Suppose that $\bar{y}(\mathbf{x})$ denotes the group average of the $N(\mathbf{x})$ values of y for which the explanatory variable vector is equal to $\mathbf{x}$, and suppose the $N(\mathbf{x})$ values are known. If the microdata are homoscedastic with conditional variance $\sigma^2 = E[u^2|\mathbf{x}]$, the correct choice of weight is

$$w(\mathbf{x}_i) = N(\mathbf{x}_i), \quad i = 1, 2, \dots, N.$$

In this case, the weighted least squares estimate of the parameter vector using the group averages is identical to the OLS estimate of the parameter vectors using the original microdata. This is the analog in the sample of the result in Problem 2.9 in W. With this estimation, one does not obtain the standard errors corresponding to the OLS estimation with the original microdata, however.